Theory of Computation-PROBLEM SET 2

March 17, 2026

Due April 2nd. Upload to learn.tsinghua.edu.cn.

Problem 1

Let $L = \{w : w \in \{0,1\}^*, w \text{ is a palindrome, and } w \text{ contains fewer 0s than 1s}\}$. Prove that L is *not* context-free. (You can see context-free languages are not closed under intersection.)

Problem 2

We have shown in class that a language L is enumerable if and only if L is Turing-recognizable. Show that L is enumerable *in strictly increasing order, with no repeats*, if and only if L is recursive (Turing-decidable).

Problem 3

Are the following languages decidable? Prove your answer.

(a) $S = \{\langle M \rangle : M \text{ is a DFA that accepts } w^R \text{ whenever it accepts } w\}$

(b) $T = \{\langle M \rangle : M \text{ is a Turing machine that accepts } w^R \text{ whenever it accepts } w\}$

Problem 4

If $A, B \subseteq \Sigma^*$ are disjoint languages (that is, $A \cap B = \emptyset$), we say that a language C **separates** A from B provided that

$$A \subseteq C \text{ and } B \cap C = \emptyset.$$

Show that, if A and B are disjoint languages such that A and B are co-recognizable, then there is a decidable languages that separates A from B.

Problem 5

Is the following language decidable? Prove your answer.

$$L = \{\langle M \rangle : M(x) \text{ is a Turing machine with running time bounded by } 100|x|^2 + 200\}$$

Problem 6 (Optional)

Prove that the language $L = \left\{x\#y : x, y \in \{0,1\}^*, x \neq y\right\}$ is context-free.